

DSA0503 – Query Processing for Data Science

Activity using SQLite Database

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1. Department Table (Parent Table) with attributes Department_ID

INTEGER PRIMARY KEY, Department_Name TEXT

The screenshot shows the DB Browser for SQLite interface. The main window displays the SQL query `SELECT * FROM Department;` and the results of the query. The results are shown in a table with two columns: Department_ID and Department_Name. The data is as follows:

Department_ID	Department_Name
101	CSE
102	ECE
103	EEE
104	AI&DS

Execution finished without errors.
Result: 4 rows returned in 11ms
At line 1:
syntax = FROM Department;

2. The Student Table (Child Table)

The screenshot shows the DB Browser for SQLite interface. The main window displays the SQL query `SELECT * FROM Student;` and the results of the query. The results are shown in a table with five columns: Student_ID, Name, Age, Department_ID, and CGPA. The data is as follows:

Student_ID	Name	Age	Department_ID	CGPA
1	Arun	20	101	8.9
2	Priya	21	102	7.6
3	Rahul	19	101	9.1
4	Sneha	20	104	8.4
5	Ravin	22	103	7.5

Execution finished without errors.
Result: 5 rows returned in 18ms
At line 1:
syntax = FROM Student;

3. Displayed students from CSE department.

The screenshot shows the DB Browser for SQLite interface. The SQL editor contains the following query:

```
1 SELECT Student.Name,  
2       Student.Student_ID,  
3       Student.Age,  
4       Student.CGPA,  
5       Department.Department_Name  
6 FROM Student  
7 JOIN Department  
8 ON Student.Department_ID = Department.Department_ID  
9 WHERE Department.Department_Name = 'CSE';
```

The results pane displays the following data:

	Name	Student_ID	Age	CGPA	Department_Name
1	Arun	1	20	8.9	CSE
2	Rahul	3	19	9.1	CSE

Execution finished without errors.
Result: 2 rows returned in 11ms
At line 1:
SELECT Student.Name,
 Student.Student_ID,
 Student.Age,
 Student.CGPA,
 Department.Department_Name
FROM Student
JOIN Department
ON Student.Department_ID = Department.Department_ID
WHERE Department.Department_Name = 'CSE';

The right sidebar shows the 'Edit Database Cell' window with 'Mode: Text' and 'Type: NULL; Size: 0 bytes'. The 'Remote' section shows 'DBHub.io' as the selected identity.

4.Counted students in each department.

The screenshot shows the DB Browser for SQLite interface. The SQL editor contains the following query:

```
1 SELECT Department.Department_Name,  
2       COUNT(Student.Student_ID) AS Total_Students  
3 FROM Department  
4 LEFT JOIN Student  
5 ON Department.Department_ID = Student.Department_ID  
6 GROUP BY Department.Department_Name;
```

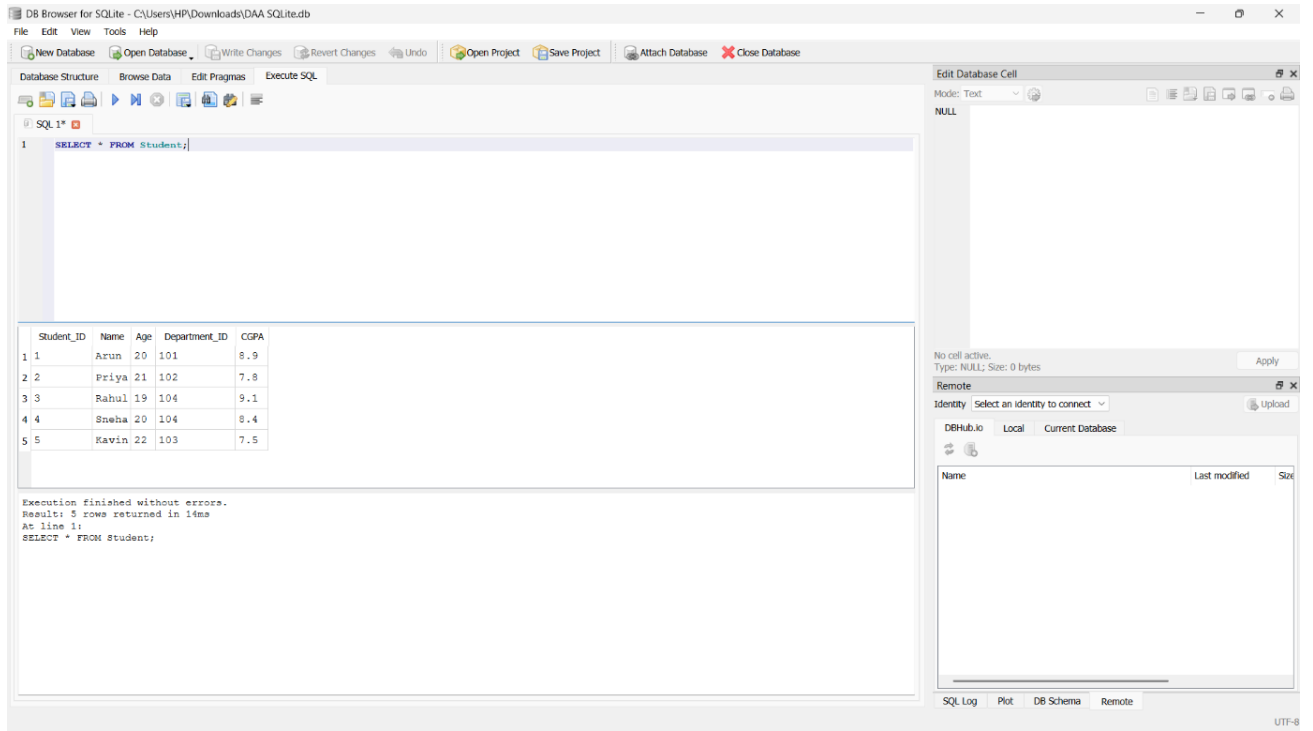
The results pane displays the following data:

	Department_Name	Total_Students
1	AIADS	1
2	CSE	2
3	ECE	1
4	EEE	1

Execution finished without errors.
Result: 4 rows returned in 11ms
At line 1:
SELECT Department.Department_Name,
 COUNT(Student.Student_ID) AS Total_Students
FROM Department
LEFT JOIN Student
ON Department.Department_ID = Student.Department_ID
GROUP BY Department.Department_Name;

The right sidebar shows the 'Edit Database Cell' window with 'Mode: Text' and 'Type: NULL; Size: 0 bytes'. The 'Remote' section shows 'DBHub.io' as the selected identity.

5. Updated the foreign key.

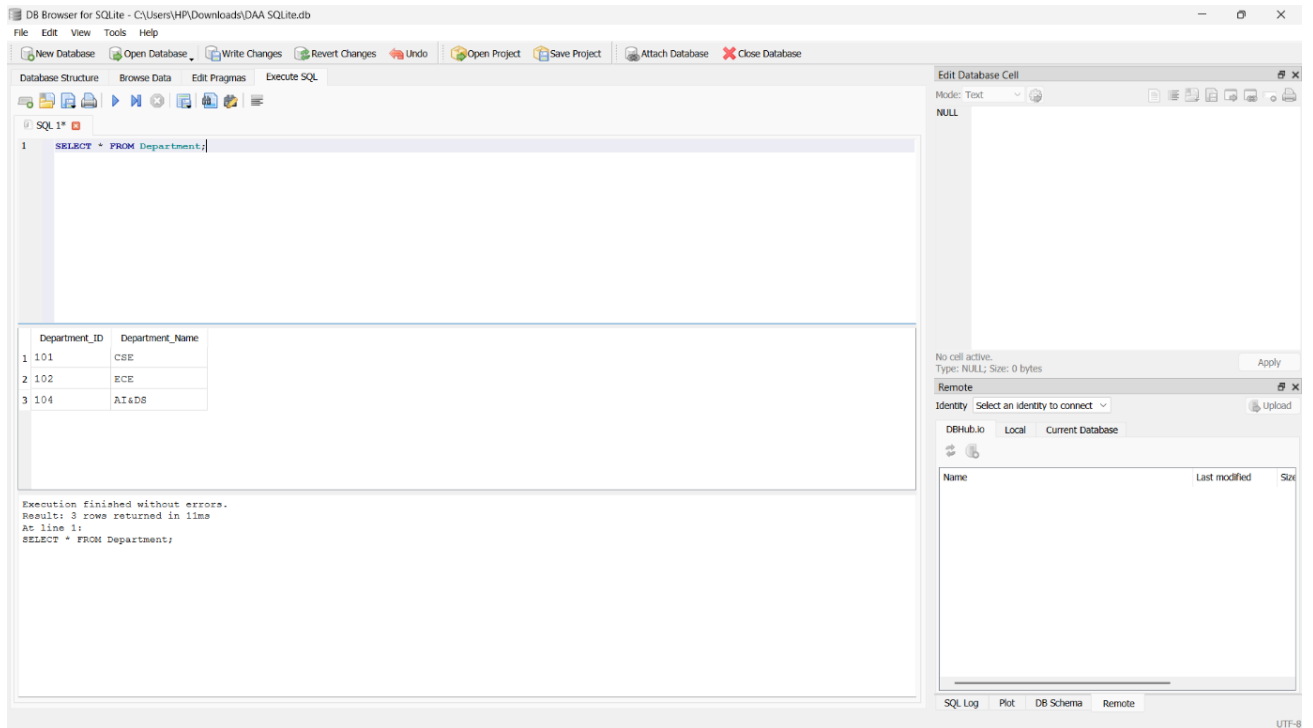


The screenshot shows the DB Browser for SQLite interface. The SQL editor contains the query `SELECT * FROM Student;`. The results pane displays a table with 5 rows and 5 columns: Student_ID, Name, Age, Department_ID, and CGPA.

Student_ID	Name	Age	Department_ID	CGPA
1	Arun	20	101	8.9
2	Priya	21	102	7.8
3	Rahul	19	104	9.1
4	Sneha	20	104	8.4
5	Kavin	22	103	7.5

Execution finished without errors.
Result: 5 rows returned in 14ms
At line 1:
`SELECT * FROM Student;`

6. Deleted a department



The screenshot shows the DB Browser for SQLite interface. The SQL editor contains the query `SELECT * FROM Department;`. The results pane displays a table with 3 rows and 2 columns: Department_ID and Department_Name.

Department_ID	Department_Name
1 101	CSE
2 102	ECE
3 104	AI&DS

Execution finished without errors.
Result: 3 rows returned in 11ms
At line 1:
`SELECT * FROM Department;`

7. Displayed the student's name, student's department and CGPA if the student's CGPA is greater than 8.

The screenshot shows the DB Browser for SQLite application. The main window displays an SQL query in the SQL editor, which has been executed. The results are shown in a table below the editor. The table has three columns: Name, Department_Name, and CGPA. The results are as follows:

	Name	Department_Name	CGPA
1	Arun	CSE	8.9
2	Rahul	AI&DS	9.1
3	Sneha	AI&DS	8.4

Below the table, the execution status is displayed: "Execution finished without errors. Result: 3 rows returned in 11ms. At line 1: SELECT Student.Name, Department.Department_Name, Student.CGPA FROM Student JOIN Department ON Student.Department_ID = Department.Department_ID WHERE Student.CGPA > 8;".

The right sidebar shows the "Edit Database Cell" window, which is currently empty. The bottom status bar indicates "UTF-8".